Physical Chemistry CUME Dec. 6th, 2013

Read "Non-equilibrium Equality for Free Energy Differences", C. Jarzynski,

Phys. Rev. Lett., **78**, *2690* and answer the following questions:

A. A 1 kg block is dragged up against gravity by a pulley system in dense liquid. The block is raised by 1 meter over 2 seconds and the pulley system did about 12 J of work. What is the reversible work during this process? What is the dissipative work in this process? (Please use a gravitational constant *g* of 9.8N/kg for this calculation).

B. Assume the same block (as described in A) is raised by 1 meter over 200 seconds, the amount of work done by the pulley system is W. Do you expect W to be greater, equal, or less than 12 J? You need to explain your reasoning to receive credit.

C. The unfolding of a protein is performed by atomic force microscopy (AFM) at 300K. A total of 4 repetitions were attempted and work performed by the AFM tip to unfold the protein is 1.00 kcal/mol, 1.20 kcal/mol, 1.25 kcal/mol, and 1.40 kcal/mol for these 4 attempts.

Please use the Jarzynski's identity to calculate the free energy of the unfolding of the protein.

Please use the second order cumulant expansion method to calculate the unfolding free energy of the protein.

D. Four research groups are trying to verify the Jarzynski's identity experimentally. Each of these groups has expertise for one type of systems. The group led by Alex use AFM to extend a polymer at 200 K, the typically fluctuation in work value associated with the extensions is 5 kcal/mol. The group led by Bob measures the work associated with implanting a nanoparticle into red blood cell. The experiments are typically done at 310K, with a fluctuation in work value of 10 kcal/mol. The group led by Charlie measures the work related with moving Au particles on a heterogeneous surface. The experiments are usually done at 2000 K and the fluctuation in work value is generally around 20 kcal/mol. The group led by David measures the delivery of a drug through a protein channel; the fluctuation in work values he measures is typically around 2 kcal/mol, and the experiments are typically performed at 300 K. Which of the four groups has the best chance of successfully verifying the Jarzynski's equality? Give me your reasoning.

E. Which of the following statement is correct according to the paper?

A. In a computer simulation, a physical heat reservoir can be simulated with the Nosé-Hoover thermostat.
B. The derivation of Jarzynski's identity is rigorously correct for modeling of a protein in a glassy liquid, where the residues of the protein are interacting strongly with the liquid molecules.
C. The Jarzynski's equality is not suitable for estimating the free energy of disintegration of a bacteria cell. The cell is fairly large and is easily visible in a $200 optical microscope.
D. The Jarzynski's equality breaks down when the switching from configuration A to configuration B is performed in one step.